OPENSOURCESDRLAB DOCUMENTATION

WaveSentry Schematic

Product	WaveSentry
Document	OSSDRLAB-WaveSentry-SCH
Version	1.0
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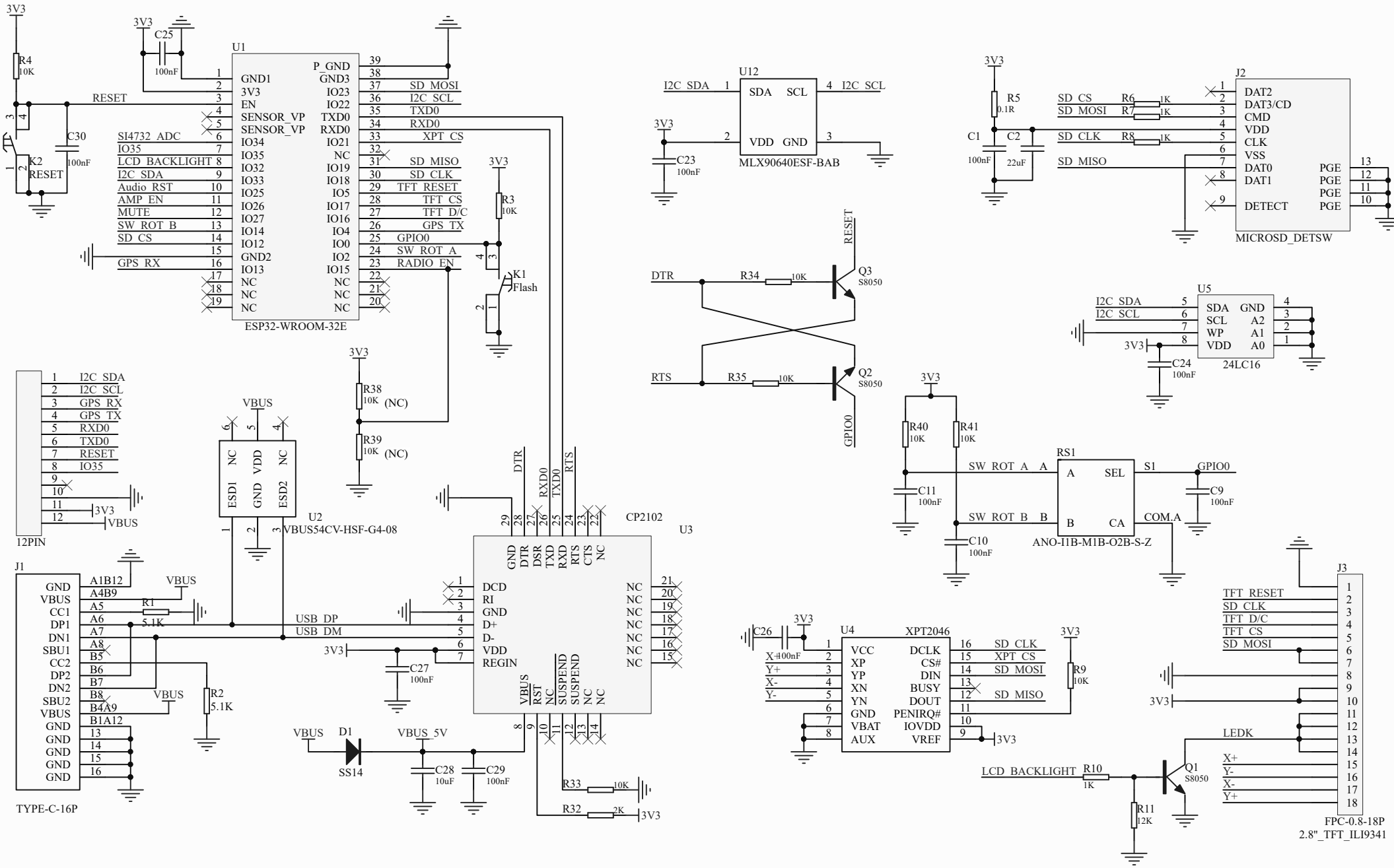
This document contains the circuit schematic of the OpenSourceSDRLab WaveSentry and WaveSentry Pro (two A4 sheets). The sheets follow this cover page.

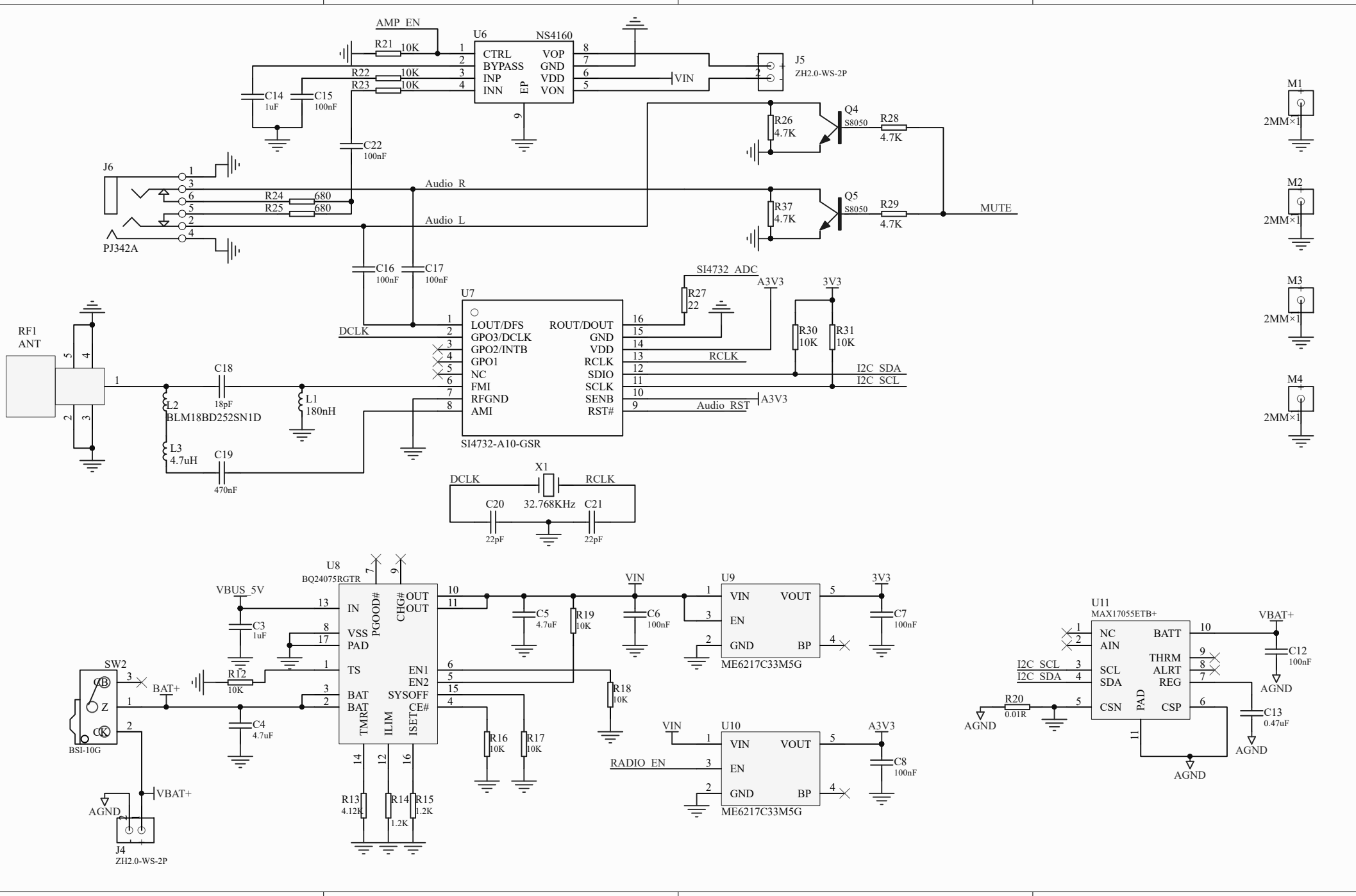
Sheet	Content
1	MCU: ESP32-WROOM-32E module, USB-C with CP2102 USB-UART bridge and auto-reset, 2.8-inch ILI9341 TFT connector with XPT2046 touch controller, microSD socket, rotary encoder and buttons, MLX90640 thermal camera (WaveSentry Pro), 24LC16 EEPROM, 12-pin expansion header (I2C, UART, GPS)
2	Audio, radio and power: SI4732 AM/FM receiver with antenna input, NS4160 audio amplifier and speaker connector, 3.5 mm headphone jack with mute switch, BQ24075 battery charger, MAX17055 fuel gauge, ME6217 3.3 V regulators, power switch and battery connector

Product page: https://opensourcesdrlab.com/products/aifw-wavesentry-esp32

Questions about the hardware: support@opensourcesdrlab.com

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U1
ESP32-WROOM-32E
GND1
3V3
EN
SENSOR_VP
SENSOR_VP
IO34
IO35
IO32
IO33
IO25
IO26
IO27
IO14
IO12
GND2
IO13
NC
NC
NC
P_GND
GND3
IO23
IO22
TXD0
RXD0
IO21
NC
IO19
IO18
IO5
IO17
IO16
IO4
IO0
IO2
IO15
NC
NC
NC
SD_MOSI
I2C_SCL
TXD0
RXD0
XPT_CS
SD_MISO
SD_CLK
TFT_RESET
TFT_CS
TFT_D/C
GPS_TX
GPIO0
SW_ROT_A
RADIO_EN
SI4732_ADC
IO35
LCD_BACKLIGHT
I2C_SDA
Audio_RST
AMP_EN
MUTE
SW_ROT_B
SD_CS
GPS_RX
RESET
K2
RESET
C30
100nF
C25
100nF
R4
10K
R3
10K
K1
Flash
U12
MLX90640ESF-BAB
SDA
SCL
VDD
GND
I2C_SDA
I2C_SCL
C23
100nF
J2
MICROSD_DETSW
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DETECT
PGE
SD_CS
SD_MOSI
SD_CLK
SD_MISO
R5
0.1R
R6
R7
R8
1K
C1
100nF
C2
22uF
U5
24LC16
SDA
SCL
WP
VDD
GND
A2
A1
A0
C24
100nF
Q3
S8050
Q2
S8050
R34
R35
10K
DTR
RTS
RESET
GPIO0
R38
R39
10K
(NC)
12PIN
I2C_SDA
I2C_SCL
GPS_RX
GPS_TX
RXD0
TXD0
RESET
IO35
3V3
VBUS
U2
VBUS54CV-HSF-G4-08
ESD1
NC
VDD
ESD2
GND
U3
CP2102
GND
DTR
DSR
TXD
RXD
RTS
CTS
NC
DCD
RI
GND
D+
D-
VDD
REGIN
VBUS
RST
SUSPEND
R40
R41
10K
RS1
ANO-I1B-M1B-O2B-S-Z
SW_ROT_A
SW_ROT_B
A
SEL
S1
B
CA
COM.A
C11
C10
C9
100nF
J1
TYPE-C-16P
GND
VBUS
CC1
DP1
DN1
SBU1
CC2
DP2
DN2
SBU2
A1B12
A4B9
A5
A6
A7
A8
B5
B6
B7
B8
B4A9
B1A12
R1
5.1K
R2
5.1K
USB_DP
USB_DM
C27
100nF
D1
SS14
VBUS_5V
C28
10uF
C29
100nF
R33
10K
R32
2K
U4
XPT2046
VCC
XP
YP
XN
YN
GND
VBAT
AUX
DCLK
CS#
DIN
BUSY
DOUT
PENIRQ#
IOVDD
VREF
C26
100nF
R9
10K
R10
1K
R11
12K
Q1
S8050
LCD_BACKLIGHT
LEDK
J3
FPC-0.8-18P
2.8"_TFT_ILI9341
TFT_RESET
SD_CLK
TFT_D/C
TFT_CS
SD_MOSI
X+
Y-
X-
Y+


AMP_EN
U6
NS4160
CTRL
BYPASS
INP
INN
VOP
GND
VDD
VON
EP
R21 10K
R22 10K
R23 10K
C14 1uF
C15 100nF
C22 100nF
J5
ZH2.0-WS-2P
VIN
J6
PJ342A
R24 680
R25 680
Audio_R
Audio_L
C16 100nF
C17 100nF
Q4
S8050
R26 4.7K
R28 4.7K
Q5
S8050
R37 4.7K
R29 4.7K
MUTE
M1
M2
M3
M4
2MM×1
U7
SI4732-A10-GSR
LOUT/DFS
GPO3/DCLK
GPO2/INTB
GPO1
NC
FMI
RFGND
AMI
ROUT/DOUT
GND
VDD
RCLK
SDIO
SCLK
SENB
RST#
DCLK
SI4732_ADC
A3V3
3V3
R27 22
R30 10K
R31 10K
RCLK
I2C_SDA
I2C_SCL
Audio_RST
RF1
ANT
C18 18pF
L2 BLM18BD252SN1D
L1 180nH
L3 4.7uH
C19 470nF
X1
32.768KHz
C20 22pF
C21 22pF
U8
BQ24075RGTR
VBUS_5V
C3 1uF
R12 10K
C4 4.7uF
SW2
BSI-10G
BAT+
VBAT+
AGND
J4
ZH2.0-WS-2P
IN
VSS
PAD
TS
BAT
BAT
TMR
ILIM
ISET
PGOOD#
CHG#
OUT
OUT
EN1
EN2
SYSOFF
CE#
C5 4.7uF
R19 10K
R18 10K
R16 10K
R17 10K
R13 4.12K
R14 1.2K
R15 1.2K
U9
ME6217C33M5G
VIN
VOUT
EN
GND
BP
C6 100nF
C7 100nF
U10
ME6217C33M5G
RADIO_EN
C8 100nF
U11
MAX17055ETB+
NC
AIN
SCL
SDA
CSN
BATT
THRM
ALRT
REG
CSP
PAD
R20 0.01R
C12 100nF
C13 0.47uF